Lesson 48: TD(λ) and Eligibility Traces

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 References **3**

3 The λ-Return: Forward View **3**
3.1 Special Cases 3

4 The λ-Return Decomposition **4**

5 Eligibility Traces and the Backward View **5**

6 Forward–Backward Equivalence **6**

7 Convergence and Trace Variants **8**
7.1 Convergence of TD(λ) 8
7.2 Trace Variants 8

8 Exercises **9**

9 Summary **10**

10 Further Reading **10**

1 Motivation and Overview

In Lesson 47 we showed that n-step TD prediction interpolates between TD(0) and Monte Carlo: small n yields low variance but high bias, while large n yields low bias but high variance. Choosing the "right" n requires prior knowledge of the problem — and the optimal n may differ across states and time steps.

TD(λ) resolves this by averaging *all* n-step returns simultaneously with geometrically decaying weights. The parameter $\lambda \in [0, 1]$ smoothly controls the bias–variance tradeoff: $\lambda = 0$ recovers TD(0), $\lambda = 1$ recovers MC, and intermediate values blend all horizons. The resulting λ-return admits a remarkable decomposition as an exponentially weighted sum of TD errors, which leads to the concept of *eligibility traces* — a memory mechanism that allows incremental, $O(|\mathcal{S}|)$-per-step computation of what would otherwise require the entire future trajectory. The tools developed in this lesson will let us prove the exact equivalence between the forward view (the λ-return) and the backward view (eligibility traces), unifying the theoretical and computational perspectives on TD learning.

Where TD(λ) and Eligibility Traces Appear

- **Generalized Advantage Estimation (GAE)** uses the λ-weighted TD error sum $\hat{A}_t = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}$ as a low-variance advantage estimator in PPO and TRPO.
- **SARSA(λ) and Q(λ)** extend eligibility traces to the control setting, propagating TD errors backward through recently visited state–action pairs.
- **True online TD(λ)** with Dutch traces achieves exact equivalence with the online λ-return algorithm, the gold standard for incremental multi-step learning.
- **Actor–critic methods** use eligibility traces in both the critic (for value estimation) and the actor (for policy gradient estimation).

Prerequisites. TD(0) update rule, TD error δ_t, unbiasedness of the TD error under V^π (Lesson 47, Lemma 4.2), the telescoping decomposition $G_t - V^\pi(S_t) = \sum_{k=0}^{\infty} \gamma^k \delta^\pi_{t+k}$ (Lesson 47, Lemma 4.4), n-step return and its telescoping decomposition (Lesson 47, Lemma 6.3). Stochastic approximation convergence theorem (Lesson 31). Absolute convergence and interchange of summation for double series.

Learning objectives.

- *Define* the λ-return and *verify* that $\lambda = 0$ gives the TD(0) target and $\lambda = 1$ gives the MC return.
- *Prove* the λ-return decomposition theorem: the λ-return minus the current value equals a sum of $(\gamma\lambda)^k$-weighted TD errors.
- *Define* the accumulating eligibility trace and the TD(λ) backward view update.
- *Prove* the forward–backward equivalence theorem in the offline setting by unrolling traces, interchanging summation, and identifying the λ-return.
- *Explain* the computational advantage of the backward view and *compare* trace variants (accumulating, replacing, Dutch).
- *State* the convergence guarantee for TD(λ) and identify the conditions under which it holds.

2 References

- Sutton and Barto [2018, Chapter 12] — Eligibility traces, forward and backward views, TD(λ), and trace variants (accumulating, replacing, Dutch).
- Sutton [1988, Section 3] — The original formulation of TD(λ) and the forward view.
- Bertsekas and Tsitsiklis [1996, Chapter 5] — Convergence analysis of TD(λ) via stochastic approximation.
- Szepesvári [2010, Section 4.3] — Compact treatment of λ-returns and their convergence properties.
- Tsitsiklis and Van Roy [1997] — Analysis of TD(λ) with linear function approximation, including convergence and error bounds.

3 The λ-Return: Forward View

Rather than selecting a single n for the n-step return, TD(λ) takes a geometrically weighted average of *all* n-step returns.

Definition 3.1 (λ-Return)**.** For $\lambda \in [0, 1]$, the *λ-return* from time t is

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}, \tag{1}$$

where $G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(S_{t+n})$ is the n-step return (Lesson 47, Definition 6.1). The normalization factor $(1-\lambda)$ ensures the weights sum to one: $(1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} = 1$ for $\lambda \in [0, 1)$.

Remark 3.2 (Geometric Weighting). The weight assigned to $G_t^{(n)}$ is $(1-\lambda)\lambda^{n-1}$, which decays geometrically in n. Short-horizon returns receive more weight when λ is small; long-horizon returns dominate when λ is close to 1. The geometric weighting is not merely convenient — it is the unique memoryless distribution on the positive integers, which leads to the recursive eligibility trace structure in the backward view.

3.1 Special Cases

Proposition 3.3 (Special Cases of the λ-Return)**.**

1. *When $\lambda = 0$: $G_t^0 = G_t^{(1)} = R_{t+1} + \gamma V(S_{t+1})$ (the TD(0) target).*
2. *When $\lambda = 1$: $G_t^1 = G_t$ (the full MC return).*

Proof. **Case $\lambda = 0$.** By Definition 3.1:

$$G_t^0 = (1-0)\sum_{n=1}^{\infty} 0^{n-1} G_t^{(n)} = 1 \cdot G_t^{(1)} = R_{t+1} + \gamma V(S_{t+1}). \tag{2}$$

Case $\lambda = 1$. We use the telescoping decomposition from Lesson 47 (Lemma 6.3): $G_t^{(n)} -$

$V(S_t) = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$. Then:

$$\begin{aligned} G_t^1 - V(S_t) &= \lim_{\lambda \to 1^-} (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \big[G_t^{(n)} - V(S_t)\big] \\ &= \lim_{\lambda \to 1^-} (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \sum_{k=0}^{n-1} \gamma^k \delta_{t+k} \\ &= \lim_{\lambda \to 1^-} \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k} = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} = G_t - V(S_t), \end{aligned} \tag{3}$$

where the second-to-last equality uses the λ-return decomposition (proved below in Theorem 4.1) and the final equality is the infinite telescoping decomposition (Lesson 47, Lemma 4.4) applied with $V = V^\pi$, together with $\lim_{n\to\infty} \gamma^n V(S_{t+n}) = 0$. Thus $G_t^1 = G_t$. □

Example 3.4 (λ-Return Interpolation)**.** Consider a trajectory with TD errors $\delta_t = 2, \delta_{t+1} = -1, \delta_{t+2} = 3$ and $\gamma = 0.9$. For different values of λ:

$$\begin{aligned} \lambda = 0: &\quad G_t^0 - V(S_t) = \delta_t = 2, \\ \lambda = 0.5: &\quad G_t^{0.5} - V(S_t) = 2 + (0.45)(-1) + (0.45)^2 \cdot 3 + \cdots \approx 2.16, \\ \lambda = 1: &\quad G_t^1 - V(S_t) = 2 + 0.9(-1) + 0.81(3) + \cdots = 3.53 + \cdots \end{aligned}$$

Higher λ incorporates TD errors from further into the future.

4 The λ-Return Decomposition

The central analytical result for the λ-return expresses it as an exponentially weighted sum of TD errors.

λ-Return Decomposition

Theorem 4.1 (λ-Return Decomposition)**.** *For $\lambda \in [0, 1)$, the λ-return satisfies*

$$G_t^\lambda - V(S_t) = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}. \tag{4}$$

Proof. Starting from Definition 3.1 and applying the n-step telescoping decomposition (Lesson 47, Lemma 6.3: $G_t^{(n)} - V(S_t) = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$):

$$\begin{aligned} G_t^\lambda - V(S_t) &= (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \big[G_t^{(n)} - V(S_t)\big] \\ &= (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}. \end{aligned} \tag{5}$$

We now interchange the order of summation. The double sum ranges over pairs (n, k) with $n \geq 1$ and $0 \leq k \leq n-1$. For a fixed $k \geq 0$, the index n ranges over $n \geq k+1$. The interchange is justified by absolute convergence: since $|\delta_{t+k}|$ is bounded (by $R_{\max}(1+\gamma)/(1-\gamma)$ on a finite MDP) and $\gamma\lambda < 1$, the double sum converges absolutely.

Interchanging:

$$G_t^\lambda - V(S_t) = (1-\lambda) \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} \sum_{n=k+1}^{\infty} \lambda^{n-1}. \tag{6}$$

Evaluate the inner geometric series. With the substitution $m = n - k - 1$ (so $n = m + k + 1$ and $\lambda^{n-1} = \lambda^{m+k}$):

$$\sum_{n=k+1}^{\infty} \lambda^{n-1} = \sum_{m=0}^{\infty} \lambda^{m+k} = \lambda^k \sum_{m=0}^{\infty} \lambda^m = \frac{\lambda^k}{1-\lambda}. \tag{7}$$

Substituting (7) into (6):

$$G_t^\lambda - V(S_t) = (1-\lambda) \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} \cdot \frac{\lambda^k}{1-\lambda} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}. \tag{8}$$

This completes the proof. □

Remark 4.2 (Interpretation). Theorem 4.1 reveals the structure of the λ-return: the prediction error $G_t^\lambda - V(S_t)$ is a weighted sum of TD errors with exponentially decaying weights $(\gamma\lambda)^k$. The parameter λ controls the effective horizon:

- $\lambda = 0$: Only the immediate TD error δ_t contributes (TD(0)).
- $\lambda = 1$: All TD errors contribute with weights γ^k (MC return, by Lesson 47, Lemma 4.4).
- $0 < \lambda < 1$: The effective number of contributing TD errors is approximately $1/(1-\gamma\lambda)$.

This is the *forward view*: at each time step, we look forward at the sequence of future TD errors.

Example 4.3 (Finite Episode)**.** For an episode of length T (with $V(S_T) = 0$), the decomposition (4) truncates to

$$G_t^\lambda - V(S_t) = \sum_{k=0}^{T-1-t} (\gamma\lambda)^k \delta_{t+k}. \tag{9}$$

For $T = 3$, $t = 0$: $G_0^\lambda - V(S_0) = \delta_0 + \gamma\lambda\,\delta_1 + (\gamma\lambda)^2\delta_2$.

The forward view requires waiting until the end of the episode to compute G_t^λ. We now develop the backward view, which achieves the same result incrementally.

5 Eligibility Traces and the Backward View

Definition 5.1 (Accumulating Eligibility Trace)**.** The *accumulating eligibility trace* for state s is defined recursively:

$$e_0(s) = 0, \qquad e_t(s) = \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\}, \tag{10}$$

where $\mathbf{1}\{S_t = s\}$ is the indicator that equals 1 if $S_t = s$ and 0 otherwise.

Remark 5.2 (Trace Intuition). The trace $e_t(s)$ records a "fading memory" of recent visits to state s. Each visit increments the trace by 1; between visits, the trace decays by $\gamma\lambda$ per step. A state visited recently and frequently has a high trace value. When a TD error δ_t occurs, credit is assigned to each state proportionally to its current trace — this is a form of temporal credit assignment.

Definition 5.3 (TD(λ) Backward View Update)**.** The *TD(λ) backward view* updates all states simultaneously at each step:

$$V(s) \leftarrow V(s) + \alpha\,\delta_t\, e_t(s) \quad \text{for all } s \in \mathcal{S}, \tag{11}$$

where $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$ is the current TD error and $e_t(s)$ is defined by (10).

Lemma 5.4 (Explicit Form of the Eligibility Trace)**.** *The eligibility trace admits the closed-form expression*

$$e_t(s) = \sum_{j=0}^{t} (\gamma\lambda)^{t-j} \mathbf{1}\{S_j = s\}. \tag{12}$$

Proof. By induction on t.

Base case ($t = 0$): $e_0(s) = \mathbf{1}\{S_0 = s\}$ from (10) (since $e_{-1}(s) = 0$ by convention, we get $e_0(s) = 0 + \mathbf{1}\{S_0 = s\}$). The formula (12) gives $\sum_{j=0}^{0}(\gamma\lambda)^{0-0}\mathbf{1}\{S_0 = s\} = \mathbf{1}\{S_0 = s\}$. ✓

Inductive step. Assume (12) holds for $t-1$. Then:

$$\begin{aligned} e_t(s) &= \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\} \\ &= \gamma\lambda \sum_{j=0}^{t-1}(\gamma\lambda)^{t-1-j}\mathbf{1}\{S_j = s\} + \mathbf{1}\{S_t = s\} \\ &= \sum_{j=0}^{t-1}(\gamma\lambda)^{t-j}\mathbf{1}\{S_j = s\} + (\gamma\lambda)^0\mathbf{1}\{S_t = s\} \\ &= \sum_{j=0}^{t}(\gamma\lambda)^{t-j}\mathbf{1}\{S_j = s\}. \end{aligned}$$

□

Example 5.5 (Trace Evolution)**.** Consider states $\{A, B, C\}$ with $\gamma\lambda = 0.8$ and the trajectory $S_0 = A, S_1 = B, S_2 = A, S_3 = C$. The trace for state A evolves as:

$$\begin{aligned} &e_0(A) = 1, \\ &e_1(A) = 0.8 \cdot 1 + 0 = 0.8, \\ &e_2(A) = 0.8 \cdot 0.8 + 1 = 1.64, \\ &e_3(A) = 0.8 \cdot 1.64 + 0 = 1.312. \end{aligned}$$

At time $t = 2$, the revisit to A causes the trace to jump above 1, reflecting the accumulated credit from two visits.

6 Forward–Backward Equivalence

We now prove the central theorem connecting the forward view (Section 3) and the backward view (Section 5).

Forward–Backward Equivalence

Theorem 6.1 (Equivalence of Forward and Backward Views)**.** *In the* offline *setting (all updates accumulated during an episode and applied at the episode's end), the total update to $V(s)$ from the backward view equals the total update from the forward view:*

$$\sum_{t=0}^{T-1} \alpha\, \delta_t\, e_t(s) = \sum_{t=0}^{T-1} \alpha\big[G_t^\lambda - V(S_t)\big]\mathbf{1}\{S_t = s\}, \tag{13}$$

where T is the episode length.

Proof. We prove the identity (13) by reorganizing the left-hand side through four steps: unrolling the trace, substituting, interchanging summation order, and identifying the λ-return decomposition.

Step 1: Unroll the eligibility trace. By Lemma 5.4:

$$e_t(s) = \sum_{j=0}^{t}(\gamma\lambda)^{t-j}\mathbf{1}\{S_j = s\}. \tag{14}$$

Step 2: Substitute into the backward view sum. The total backward-view update (ignoring the common factor α, which appears on both sides of (13)) is:

$$\sum_{t=0}^{T-1} \delta_t \, e_t(s) = \sum_{t=0}^{T-1} \delta_t \sum_{j=0}^{t} (\gamma\lambda)^{t-j} \mathbf{1}\{S_j = s\}. \tag{15}$$

This is a double sum over pairs (t, j) with $0 \le j \le t \le T-1$.

Step 3: Interchange the order of summation. We swap the order of summation over t and j. The summation region is $\{(t, j) : 0 \le j \le t \le T-1\}$. For a fixed j, the index t ranges from j to $T-1$:

$$\sum_{t=0}^{T-1} \delta_t \, e_t(s) = \sum_{j=0}^{T-1} \mathbf{1}\{S_j = s\} \sum_{t=j}^{T-1} (\gamma\lambda)^{t-j} \delta_t. \tag{16}$$

The interchange is valid because the double sum involves finitely many terms (both indices range over $\{0, \ldots, T-1\}$).

Step 4: Identify the forward view. In the inner sum, substitute $k = t - j$ (so $t = j + k$ and k ranges from 0 to $T - 1 - j$):

$$\sum_{t=j}^{T-1} (\gamma\lambda)^{t-j} \delta_t = \sum_{k=0}^{T-1-j} (\gamma\lambda)^k \delta_{j+k}. \tag{17}$$

By the λ-return decomposition (Theorem 4.1, applied to the finite-episode case where the sum truncates at the episode boundary), this equals $G_j^\lambda - V(S_j)$.

Substituting back into (16):

$$\sum_{t=0}^{T-1} \delta_t \, e_t(s) = \sum_{j=0}^{T-1} \mathbf{1}\{S_j = s\} \big[G_j^\lambda - V(S_j)\big]. \tag{18}$$

Multiplying both sides by α and relabeling the dummy index j as t yields (13). □

Remark 6.2 (Why "Offline"?). Theorem 6.1 holds in the *offline* setting where value estimates V remain fixed throughout the episode and all updates are accumulated. In the *online* setting, V changes after each update, so the TD errors δ_t in the backward view depend on the updated values while the forward-view G_t^λ is computed with the original values. The equivalence breaks down slightly, motivating the "true online" algorithms discussed in Section 7.2.

Computational Advantage of the Backward View

The forward view requires computing G_t^λ for each time step t, which needs future rewards up to the end of the episode — making it inherently non-causal and requiring $O(T)$ memory for the full trajectory. The backward view (11) uses only:

- The current TD error δ_t (available immediately after the transition).
- The trace vector $e_t \in \mathbb{R}^{|\mathcal{S}|}$ (updated incrementally via (10)).

The computational cost is $O(|\mathcal{S}|)$ per step (updating all traces and values), making the backward view suitable for online, incremental learning. Theorem 6.1 guarantees that both views produce identical total updates in the offline case.

Example 6.3 (Forward–Backward Equivalence Verification)**.** Consider an episode with $T = 3$, states $\{A, B\}$, and trajectory $S_0 = A, S_1 = B, S_2 = A$ with TD errors $\delta_0 = 1$, $\delta_1 = -0.5$, $\delta_2 = 2$. Let $\gamma\lambda = 0.8$.

Backward view for state A ($\alpha = 1$):

$$\begin{aligned}&\text{Traces: } e_0(A) = 1,\ e_1(A) = 0.8,\ e_2(A) = 0.8 \cdot 0.8 + 1 = 1.64.\\&\text{Total: } \delta_0 e_0(A) + \delta_1 e_1(A) + \delta_2 e_2(A) = 1(1) + (-0.5)(0.8) + 2(1.64) = 3.88.\end{aligned}$$

Forward view for state A (visited at $t = 0$ and $t = 2$):

$$\begin{aligned}&G_0^\lambda - V(S_0) = \delta_0 + 0.8\,\delta_1 + 0.64\,\delta_2 = 1 - 0.4 + 1.28 = 1.88,\\&G_2^\lambda - V(S_2) = \delta_2 = 2,\\&\qquad\text{Total: } 1.88 + 2 = 3.88. \quad \checkmark\end{aligned}$$

7 Convergence and Trace Variants

7.1 Convergence of TD(λ)

Corollary 7.1 (Convergence of TD(λ))**.** *Under the following conditions:*

1. *The MDP has finite state and action spaces.*
2. *The policy π is stationary.*
3. *Every state is visited infinitely often.*
4. *Step sizes satisfy $\sum_n \alpha_n(s) = \infty$ and $\sum_n \alpha_n^2(s) < \infty$ for each s.*

For any $\lambda \in [0, 1]$, the offline TD(λ) algorithm converges to V^π almost surely.

Proof. By Theorem 6.1, offline TD(λ) applies the same total update as the forward-view algorithm that uses G_t^λ as the target. We show that this forward-view algorithm is a valid stochastic approximation.

Define the λ-weighted Bellman operator $\mathcal{T}_\lambda^\pi \colon \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ by

$$(\mathcal{T}_\lambda^\pi V)(s) = \mathbb{E}_\pi\big[G_t^\lambda \mid S_t = s\big] = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} (\mathcal{T}^\pi)^n V(s). \tag{19}$$

This is a weighted average of the operators $(\mathcal{T}^\pi)^n$, each of which is a γ^n-contraction. The operator $\mathcal{T}_\lambda^\pi$ is therefore a contraction with rate

$$(1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \gamma^n = \frac{(1 - \lambda)\gamma}{1 - \lambda\gamma}, \tag{20}$$

which is strictly less than 1 for $\gamma < 1$. The unique fixed point is V^π, since $\mathcal{T}^\pi V^\pi = V^\pi$ implies $\mathcal{T}_\lambda^\pi V^\pi = V^\pi$.

The forward-view update is a stochastic approximation of $V \leftarrow \mathcal{T}_\lambda^\pi V$, with noise $w_t = G_t^\lambda - (\mathcal{T}_\lambda^\pi V)(S_t)$ satisfying $\mathbb{E}[w_t \mid S_t, V] = 0$ by construction. The noise variance is bounded (since G_t^λ is a convex combination of bounded n-step returns). All conditions of the Robbins–Monro theorem are satisfied, yielding $V_n(s) \to V^\pi(s)$ almost surely. □

Remark 7.2 (Contraction Rate as a Function of λ). The contraction rate (20) is $(1-\lambda)\gamma/(1-\lambda\gamma)$. For $\lambda = 0$ this equals γ (the TD(0) rate), and as $\lambda \to 1$ it approaches γ from below (since the $(1 - \lambda)$ factor in the numerator shrinks). For intermediate λ, the contraction rate is strictly smaller than γ, meaning the expected operator contracts *faster* than TD(0). However, the variance of G_t^λ increases with λ, so the faster contraction per iteration may be offset by noisier updates.

7.2 Trace Variants

Definition 7.3 (Replacing Traces)**.** The *replacing trace* caps the trace at 1 upon revisit:

$$e_t(s) = \begin{cases} 1 & \text{if } S_t = s, \\ \gamma\lambda\, e_{t-1}(s) & \text{otherwise.} \end{cases} \tag{21}$$

Equivalently: $e_t(s) = \max\big(\gamma\lambda\, e_{t-1}(s),\ \mathbf{1}\{S_t = s\}\big)$.

Remark 7.4 (Replacing vs. Accumulating). Accumulating traces can grow without bound in states that are visited repeatedly within a short time window (e.g., $e_t(s) > 1$ in Example 5.5). Replacing traces prevent this by resetting to 1 upon each visit, which can improve stability in practice. However, replacing traces do *not* satisfy the exact forward–backward equivalence of Theorem 6.1.

Definition 7.5 (Dutch Traces)**.** The *Dutch trace* is defined by

$$e_t(s) = \gamma\lambda\, e_{t-1}(s) + \alpha\big(1 - \gamma\lambda\, e_{t-1}(s)\big)\mathbf{1}\{S_t = s\}, \tag{22}$$

where α is the step size.

Remark 7.6 (True Online TD(λ)). Dutch traces are designed to produce exact equivalence with the *online* λ-return algorithm, where value estimates are updated after every step (not just at episode end). The resulting "true online TD(λ)" algorithm [Sutton and Barto, 2018, Chapter 12] computes the same sequence of value estimates as the theoretically ideal online forward-view algorithm, making it the method of choice when both computational efficiency and theoretical exactness are desired.

Accumulating Traces Break Online Equivalence

Theorem 6.1 holds only in the offline setting. When using standard accumulating traces with online updates (applying each update immediately), the resulting algorithm does *not* compute the same updates as the online forward view. The discrepancy arises because the TD errors δ_t in the backward view depend on the updated value function, while the forward-view λ-return uses the original values. For exact online equivalence, use Dutch traces.

8 Exercises

Exercise 8.1 (Normalization of the λ-Return Weights)**.** ($*$)
Verify that the weights in Definition 3.1 sum to one: show that $(1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1} = 1$ for $\lambda \in [0, 1)$. What happens when $\lambda = 1$, and how is the λ-return interpreted in this limiting case?

Exercise 8.2 (λ-Return for $\lambda = 0.5$)**.** ($*$)
Consider a 4-step episode with trajectory S_0, S_1, S_2, S_3 (terminal), rewards $R_1 = 1$, $R_2 = 0$, $R_3 = 2$, $\gamma = 0.9$, and value estimates $V(S_1) = 3$, $V(S_2) = 2$, $V(S_3) = 0$. Compute $G_0^{(1)}$, $G_0^{(2)}$, $G_0^{(3)}$, and $G_0^{0.5}$ directly from the definitions. Verify that $G_0^{0.5} - V(S_0)$ equals $\sum_{k=0}^{2}(0.45)^k\delta_k$ using the decomposition theorem.

Exercise 8.3 (Trace Unrolling for a Specific Trajectory)**.** ($**$)
Consider $\mathcal{S} = \{A, B, C\}$ with $\gamma\lambda = 0.7$ and the trajectory $S_0 = A, S_1 = B, S_2 = B, S_3 = A, S_4 = C$. Compute $e_t(s)$ for all $s \in \mathcal{S}$ and $t = 0, 1, 2, 3, 4$ using both the recursive definition (10) and the explicit formula (12). Verify they agree.

Exercise 8.4 (Forward–Backward Equivalence Proof Variant)**.** (∗∗)
Re-derive the forward–backward equivalence (Theorem 6.1) by starting from the forward view $\sum_{t=0}^{T-1}[G_t^\lambda - V(S_t)]\mathbf{1}\{S_t = s\}$ and showing it equals $\sum_{t=0}^{T-1} \delta_t\, e_t(s)$. That is, work the proof in the reverse direction.

Hint: Expand $G_t^\lambda - V(S_t)$ using the decomposition theorem, then interchange the order of summation.

Exercise 8.5 (Contraction Rate Analysis)**.** (∗∗)
For $\gamma = 0.99$, plot the contraction rate $(1-\lambda)\gamma/(1-\lambda\gamma)$ of the $\mathcal{T}_\lambda^\pi$ operator as a function of $\lambda \in [0, 1)$. Determine the value of λ that minimizes this contraction rate and compute the minimum. Explain why a faster contraction rate does not necessarily mean faster convergence in practice.

Exercise 8.6 (Replacing Traces and Forward–Backward Mismatch)**.** (∗ ∗ ∗)
Construct a concrete example (episode with 4 states, $T = 5$) where the total backward-view update with *replacing* traces differs from the forward-view update $\sum_t[G_t^\lambda - V(S_t)]\mathbf{1}\{S_t = s\}$. Quantify the discrepancy and explain which feature of the replacing trace causes the mismatch.

Hint: The mismatch arises when a state is visited multiple times in quick succession.

9 Summary

- The **λ-return** $G_t^\lambda = (1-\lambda)\sum_{n=1}^\infty \lambda^{n-1} G_t^{(n)}$ is a geometrically weighted average of all n-step returns, with $\lambda = 0$ recovering TD(0) and $\lambda = 1$ recovering MC.
- The **λ-return decomposition** (Theorem 4.1) shows that $G_t^\lambda - V(S_t) = \sum_{k=0}^\infty (\gamma\lambda)^k \delta_{t+k}$, expressing the prediction error as an exponentially weighted sum of TD errors.
- The **accumulating eligibility trace** $e_t(s) = \gamma\lambda\, e_{t-1}(s) + \mathbf{1}\{S_t = s\}$ provides a fading memory of state visits, enabling incremental $O(|\mathcal{S}|)$-per-step credit assignment.
- The **forward–backward equivalence** (Theorem 6.1) proves that the total offline update from the backward view (using traces) equals the total update from the forward view (using λ-returns), established via trace unrolling and summation interchange.
- **TD(λ) converges** to V^π almost surely under standard conditions, with the λ-weighted Bellman operator providing a contraction rate of $(1-\lambda)\gamma/(1-\lambda\gamma) < 1$.
- **Trace variants**: accumulating traces give exact offline equivalence; replacing traces cap at 1 for stability; Dutch traces achieve exact *online* equivalence with the λ-return algorithm.

10 Further Reading

- **Lesson 49** extends TD(λ) to the control setting, developing SARSA(λ) and Watkins's Q(λ) with eligibility traces for action-value estimation.
- Sutton [1988] is the foundational paper on TD(λ), establishing the forward-view formulation and early convergence results.
- Sutton and Barto [2018, Chapter 12] provides the definitive textbook treatment of eligibility traces, including the true online TD(λ) algorithm with Dutch traces and the general λ-return algorithm.
- Tsitsiklis and Van Roy [1997] analyzes TD(λ) with linear function approximation, proving convergence and characterizing the approximation error as a function of λ. This analysis shows that larger λ yields smaller asymptotic bias in the function approximation setting.

- For the connection to policy gradient methods, Generalized Advantage Estimation (GAE) uses the decomposition $\hat{A}_t^{\mathrm{GAE}(\lambda)} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}$ as a variance-reduced advantage estimator, directly applying Theorem 4.1.
- Borkar [2008] provides the stochastic approximation machinery underlying all TD convergence proofs, including extensions to the function approximation setting.

References

Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming*. Athena Scientific, 1996.

Vivek S. Borkar. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.

Richard S. Sutton. Learning to predict by the methods of temporal differences. *Machine Learning*, 3(1):9–44, 1988.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.

Csaba Szepesvári. *Algorithms for Reinforcement Learning*. Morgan & Claypool, 2010.

John N. Tsitsiklis and Benjamin Van Roy. An analysis of temporal-difference learning with function approximation. *IEEE Transactions on Automatic Control*, 42(5):674–690, 1997.